Table D.1: Confusion matrix of the classification of A-type asteroids by our algorithm. The list of 389 asteroids both classified by DeMeo et al. (2009, 2014, 2019) and having a DR3 spectrum considered for this study contains 21 A-type asteroids.

Predicted \ True	A-type	Non-A-type
A-type	15	8
Non-A-type	6	360

classified from VIS-NIR observations by DeMeo et al. (2009, 2014, 2019). This list contains 389 asteroids. We applied our classification method on their DR3 spectra, in the wavelength range from 0.45 to 0.90 µm. This wavelength range was chosen in order to not take into account the first and last two bands of the DR3 spectra often affected by systematic errors Gaia Collaboration et al. (2022); Oszkiewicz et al. (2023), and to limit the effect of a potential DR3 reddening. Moreover, a quality flag going from 0 (good quality) to 2 (poorer quality) was assigned to each of the 16 bands of the Gaia DR3 spectra. We did not consider the bands flagged with a non-zero number, as done in Galinier et al. (2023).

We studied the performances of our algorithm for the classification of A-types using the confusion matrix presented in Table D.1. In the list of 389 asteroids, 21 are classified as A-types. We found that eight asteroids that are not A-type in DeMeo et al. (2009, 2014, 2019) are classified A-type by our method. This corresponds to a false positive rate of around 2.2%. However, among these false positives, three asteroids were later classified as A-type by Mahlke et al. (2022) in his own taxonomic scheme. If we consider this information, the false positive rate goes down to less than 1.5%.

Among the 21 A-type asteroids of the list, 15 get correctly classified as A-types as first or second best match with the Bus-DeMeo templates. This corresponds to a true positive rate of above 71%. The accuracy of the classification corresponds to the sum of the true positive and the true negatives, divided by the total population. The accuracy for the A-type classification here is therefore of $\frac{15+360}{389} = 96\%$.

Among the six A-type asteroids of the literature that did not get classified as A-type here (false negative rate of 28.6%), four asteroids have a S/N<21. We conclude that the classification of low S/N A-type asteroids should be taken with more caution: it is not impossible that a low S/N asteroid that was not found A by our method is an A. Similarly, asteroids found A by our method and that show a S/N<21 might not be real A.

We did not aim to extensively study the performance of our classification algorithm on other taxonomic types. We therefore advise caution with the use of spectral matching between DR3 spectra and Bus-DeMeo templates to classify asteroids in the DR3 dataset.

Appendix E: Figures

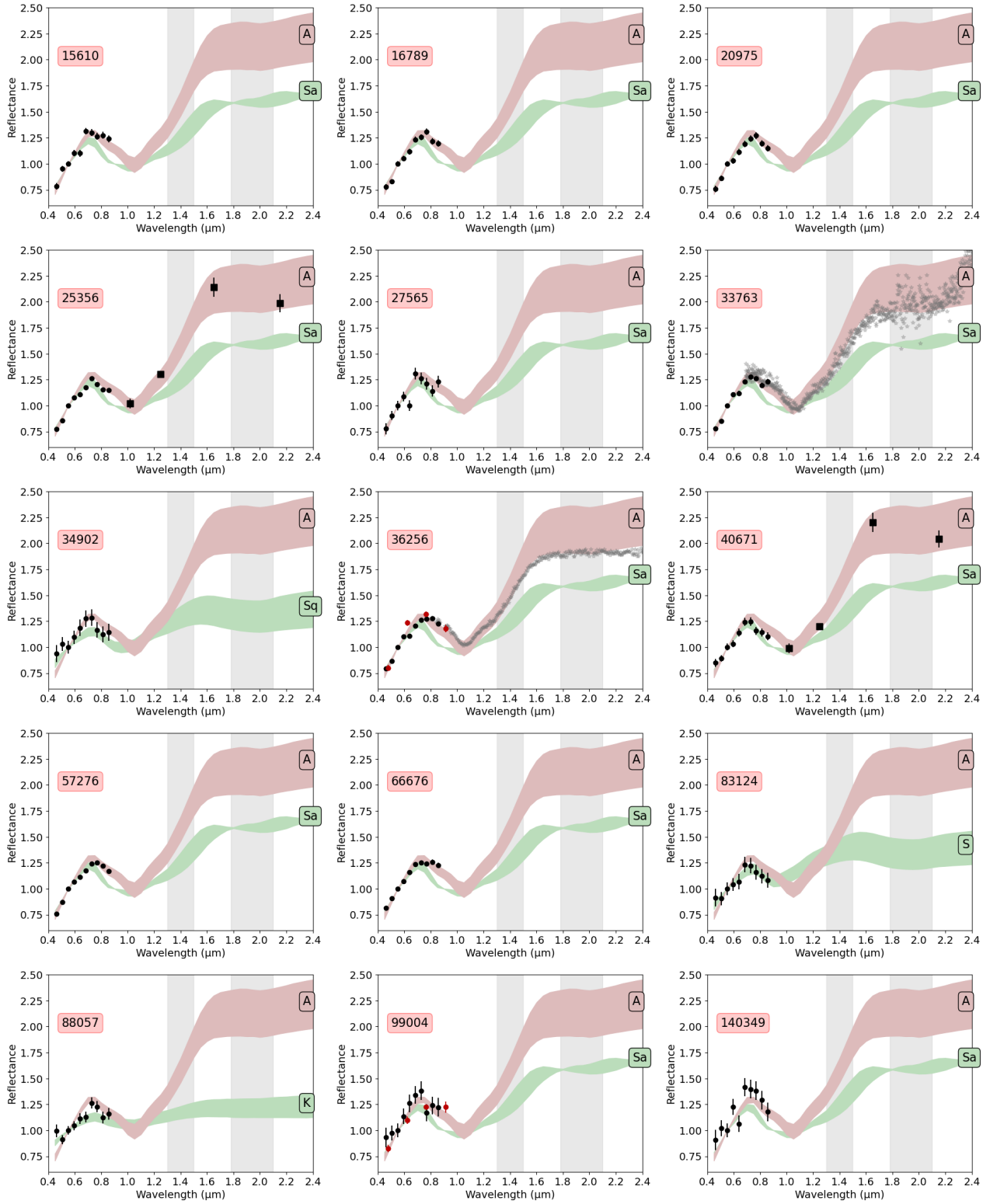
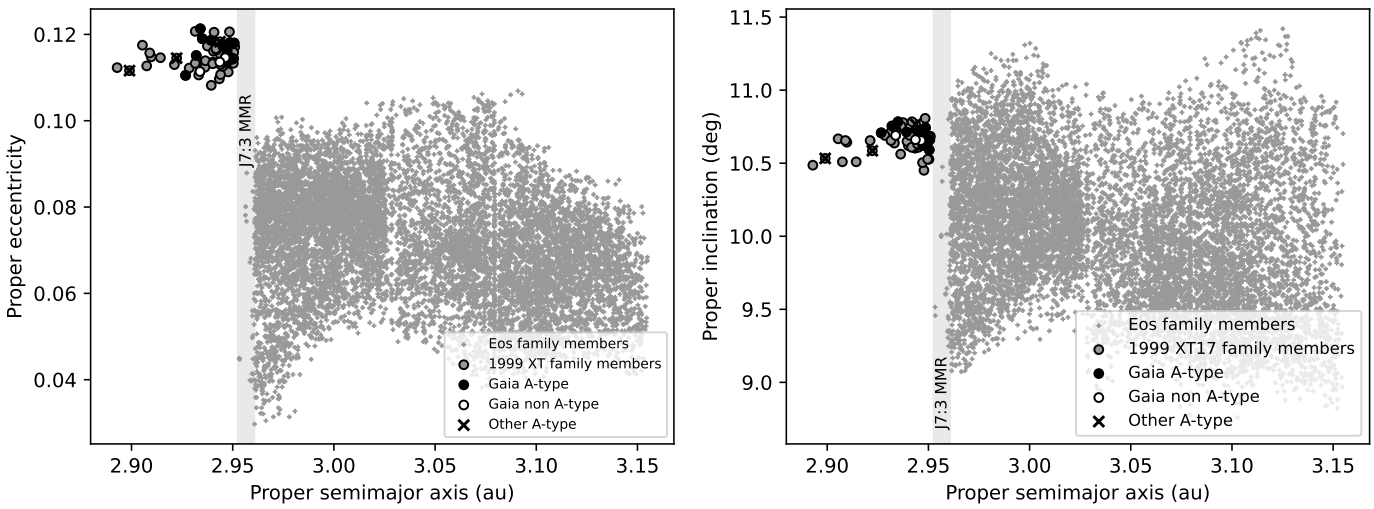
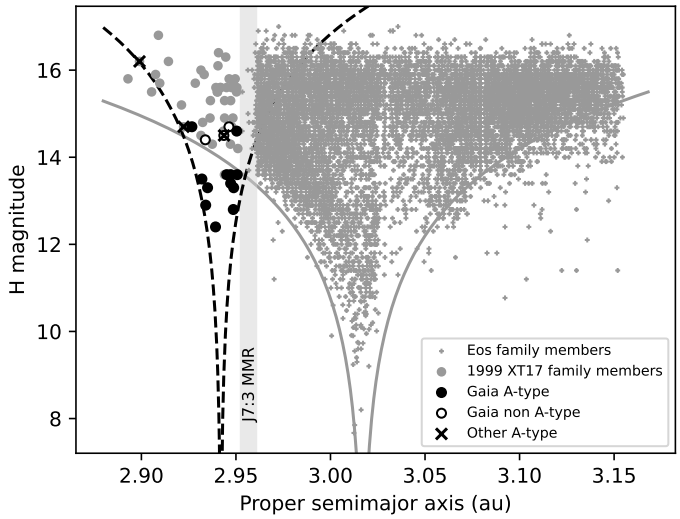


Fig. E.1: Available data of the members of the family (36256) 1999 XT17 in the Gaia DR3: Gaia spectra (black dots), SDSS data from DeMeo & Carry (2013) (red dots), MOVIS data from Popescu et al. (2018a) (black squares), and NIR spectra from DeMeo et al. (2019) (grey stars). The shaded coloured areas represent the reflectance spectra from the A-type and other classes templates of DeMeo et al. (2009).



(a) (b)



(c)

Fig. E.2: Proper orbital elements (panel a and b) and V-shape (panel c) plots of the 1999 XT17 family and the (221) Eos family. The members of Eos are displayed as grey crosses, and those of the 1999 XT17 family as circles. Black filled circles are 1999 XT17 family members that were classified as A-types by our algorithm, while the empty circles are members that were not classified as A-types. Family members that were classified A-types in the literature and have no DR3 spectra are marked as "X". Among the latter group, asteroid 76627 overlaps in the plot with asteroid 88052 (non A-type), due to their similar proper orbital elements and H magnitudes. The position of the 7:3 mean motion resonance with Jupiter is highlighted with a shade of light-grey. The V-shapes of the two families are drawn by eye following the equation $H = 5\log_{10}(|a - a_c|/C_0)$ with $a_C = 2.942$ au and $C_0 = 1.8 \times 10^{-5}$ au for the 1999 XT17 family, and $a_C = 3.017$ au and $C_0 = 2.25 \times 10^{-4}$ au for the Eos family.